

ADR-001: Medallion Architecture for NorthPeak Data Platform

Status: Accepted

Date: 14 July 2026

Author: Data Platform Engineering Team

Deciders: Aisha Okafor (Data Engineering Manager), Marcus Webb (Head of Data Platform)

Context

NorthPeak Retail Group operates five disconnected source systems from acquisitions (2018-2023):

- NP Core (PostgreSQL 14) - primary EPOS, orders, customers, inventory
- GroceryDirect (MongoDB 6.0) - online orders, returns, baskets
- SupplyLink (SQL Server 2019) - DC stock movements, purchase orders
- RewardsCo (REST API JSON) - loyalty members, points transactions
- NP Financial (Azure SQL) - FCA-regulated credit and insurance

The business requires: unified revenue reporting, GDPR DSAR within 30 days, T+1 dashboards, and a platform built with free/open source tools that maps to enterprise Databricks stacks.

Decision

Implement the **Medallion Architecture** (Bronze / Silver / Gold) using PySpark, Delta Lake OSS, DuckDB, dbt Core, and Apache Airflow.

Layers

Bronze: Raw, immutable, append-only Parquet partitioned by ingestion_date. No transformation. Full audit trail.

Silver: Cleansed Delta tables. Deduplication, PII masking (SHA-256), timezone normalisation (all UTC), null handling (UNKNOWN_CUSTOMER flag), quarantine for failed records.

Gold: Star schema via dbt. FactSales, FactInventory, FactLoyaltyPoints, DimCustomer, DimProduct, DimStore, DimDate, DimSupplier.

Alternatives Considered

Single flat warehouse: Rejected - no audit trail, cannot reprocess, unsafe PII boundary.

Fivetran + Snowflake only: Rejected - paid tools, no PySpark skills demonstrated, not accessible to candidates.

Lambda architecture (batch + streaming): Deferred to Phase 8 optional extension. Batch meets all current SLAs.

Local to Enterprise Mapping

- | Local | Enterprise Equivalent |
 - |-----|-----|
 - | PySpark local | Databricks Runtime / AWS EMR |
 - | Delta Lake OSS | Databricks Delta Lake |
 - | DuckDB | Snowflake / Azure Synapse |
 - | Airflow Docker | Databricks Workflows / AWS MWAA |
 - | GitHub Actions | Azure DevOps / Jenkins |
 - | Terraform plan | Azure / AWS full deployment |
 - | Great Expectations | Monte Carlo / Acceldata |
-

Review Date: 21 September 2026 (end of Sprint 4)